Journal of *Marine Science and Engineering*

MDPI

Review

AIS Cybersecurity: Challenges, Vulnerabilities, and Mitigation Strategies †

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† A preliminary, shorter version of this work appeared in the Proceedings of the 7th International Symposium on Cyber Security, Cryptology, and Machine Learning (CSCML) 2023, Be'er Sheva, Israel 29–30 June 2023.

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Abstract

Maritime operations rely on the Automatic Identification System (AIS), an open broadcast protocol whose unauthenticated, self-reported messages are easily abused. This survey provides an AIS-first, security-focused synthesis of AIS cybersecurity research. It makes three main contributions. First, it explains AIS protocol mechanics and uses them to derive the main security weaknesses that arise from open VHF broadcast, self-reported data, lack of built-in authentication and replay protection, and GNSS dependence. Second, it organizes AIS threats and mitigations into a unified taxonomy that links attack vectors, technical effects, operational impacts, and security measures across the prevent–detect–respond–recover lifecycle. Third, it assesses practical defenses, including authentication proposals, endpoint and network hardening, behavior-aware analytics, cross-sensor validation, and governance measures. The survey shows that existing work is strongest on anomaly detection and AIS data analytics, whereas standards-compatible authentication, scalable key management, backward-compatible deployment, trust-aware display integration, and operational validation remain open challenges. Consequently, AIS security is likely to require layered, staged, and standards-compatible mitigations rather than a single technical fix. By bringing together cybersecurity, maritime operations, and data-science perspectives, the survey provides practical guidance for securing AIS-based systems and highlights open problems for future standardization and implementation.

Keywords: automatic identification system (AIS); maritime anomaly detection; maritime cyber challenges; maritime cyber security

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Academic Editors: Rob Brown and Francis Obeng

Received: 7 May 2026
Revised: 23 June 2026
Accepted: 29 June 2026
Published: 8 July 2026

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1. Introduction

Maritime operations depend on the Automatic Identification System (AIS) for collision avoidance, traffic management, and wide-area monitoring. Yet AIS is an open VHF broadcast of unauthenticated, self-reported data. Anyone within range can listen; with the right equipment, attackers can inject, alter, or suppress messages, and AIS's reliance on Global Navigation Satellite Systems (GNSS) makes it sensitive to spoofing and jamming. These weaknesses have safety, commercial, and geopolitical consequences: they can degrade bridge situational awareness, disrupt port and logistics flows, and enable sanction evasion or misinformation [1].

Concrete AIS-specific examples illustrate these consequences. The 2019 Elba Island spoofing case shows the safety dimension: reconstructed bridge displays showed phan-

tom vessels that could create misleading collision cues and degrade bridge situational awareness [2,3]. Suspicious ship-to-ship oil-transfer cases involving falsified or missing AIS positions illustrate the commercial and law-enforcement dimension, because AIS manipulation can obscure vessel activity, cargo movements, and sanctions-relevant behavior [4]. Forged public AIS tracks placing naval vessels near contested waters illustrate the geopolitical and misinformation dimension, as manipulated AIS feeds can create false narratives about vessel presence and movement [5,6]. In addition, experimental AIS denial-of-service results show how message flooding can impair AIS availability, with potential effects on traffic monitoring and port/logistics coordination [7].

This survey takes an AIS-first, security-focused view. Drawing on AIS-security research, standards, incident reports, bridge/shore operating practice, and the detection/ML literature, we synthesize how AIS protocol mechanics give rise to security weaknesses and operational risks. Unlike prior surveys that primarily focus on AIS anomaly detection, AIS applications, or broad maritime cybersecurity, we start from AIS protocol design and operation, map the threat surface from on-air injection and GNSS spoofing/jamming to feed tampering and endpoint compromise, and link these attacks to technical effects, operational impacts, practical defenses, deployment constraints, and the remaining standardization gaps.

Maritime Cybersecurity Context

The maritime domain is a highly interconnected ecosystem that includes vessels, ports, logistics infrastructure, cargo supply chains, and shipping companies [8]. Vessel and shore systems combine information technology (IT), operational technology (OT), wireless communications, navigation systems, cargo-handling systems, and external service providers. This interconnection improves operational efficiency, but also expands the cyberattack surface and increases dependence on digital systems whose compromise may affect safety, logistics, and trade [9–11].

Sector-wide maritime cyber incidents, not specific to AIS, further illustrate the operational consequences of cyber exposure. Ransomware and destructive malware have disrupted fleet-management systems, ports, shipping companies, and global logistics, including the DNV ShipManager incident and the NotPetya attack on Maersk [12–15]. These risks are amplified by legacy technology, third-party connectivity, and uneven cybersecurity awareness and staffing.

Maritime cyber risk is addressed by a growing body of guidance and regulation. The International Maritime Organization (IMO) provides high-level guidance on maritime cyber-risk management and requires cyber risk to be addressed within Safety Management Systems through Resolution MSC.428(98) [16–18]. Complementary guidance has been issued by organizations such as the International Chamber of Shipping (ICS), BIMCO, the U.S. Coast Guard, and European regulators [19–22]. Against this broader background, AIS is a particularly important case: it is widely deployed, safety-relevant, openly broadcast, and still lacks built-in authentication and integrity protection.

The remainder of this article is organized as follows. Section 2 provides a technical overview of AIS. Section 3 characterizes AIS threats and vulnerabilities. Section 4 presents mitigation strategies grouped into protocol/authentication, endpoint/network hardening, analytics, and resilience/governance. Section 5 synthesizes the findings and outlines recommendations.

2. An Overview of the Automatic Identification System

The Automatic Identification System (AIS) is a VHF data-link system used for ship-to-ship and ship-to-shore exchange of navigation and identification data. AIS broadcasts

dynamic information such as position, course, and speed, together with identity, static, and voyage-related data such as Maritime Mobile Service Identity (MMSI), ship name, vessel type, and cargo-related information [23,24]. These data are displayed on dedicated AIS equipment or integrated into bridge systems such as the Electronic Chart Display and Information System (ECDIS) and radar overlays, supporting collision avoidance, traffic monitoring, and maritime-domain awareness.

The same design choices that make AIS widely interoperable also create its main security weaknesses: AIS messages are openly broadcast, self-reported, and unauthenticated; AIS position and timing depend on GNSS and onboard sensors; and many vessels share the same VHF channels through TDMA-based access.

2.1. Types of AIS Systems

AIS is categorized into Class A and Class B systems, alongside AIS base stations [25]. Class A, the initial standard, offers high transmission power and is mandatory for larger vessels due to its frequent position reporting and support for voyage-related data [26]. Class B provides a cost-effective solution, fostering broader adoption and significantly increasing maritime communication data volume [24]. Following Class B's introduction, regulatory frameworks, such as the European Union's 2010 mandate, required AIS on most commercial inland vessels and fishing vessels over 15 m, expanding AIS deployment [27].

2.1.1. AIS Class A

Class A equipment complies with IMO standards, transmitting position data autonomously every 2–10 s, depending on speed and course changes, or every three minutes when stationary [28]. It also transmits static and voyage-related information every six minutes, supports safety-related text messaging, and transmits Application-Specific Messages (ASMs) for meteorological and navigational data.

2.1.2. AIS Class B

Class B AIS equipment is compatible with other AIS stations but transmits at lower power and less frequently than Class A [24]. It is intended mainly for smaller vessels and does not meet all performance requirements adopted for Class A equipment. Class B position reports are transmitted using message type 18, with reporting intervals that depend on vessel status and movement.

Class B equipment can receive safety-related text messages and application-specific messages. The availability of transmission functions for such messages is equipment- and profile-dependent: some Class B implementations may support limited transmission of selected safety-related or binary messages as optional functions, subject to the applicable standard, configuration, and operational restrictions, such as manual initiation and limits on automatic repetition.

2.1.3. AIS Base Station

Shore-based AIS base stations transmit Base Station Reports using AIS message type 4, typically including base-station identity, UTC/time information, and position information for synchronization and AIS service management.[1]

2.1.4. AIS Aid to Navigation (AtoN)

AIS AtoNs, shore-based or mobile, broadcast navigational-aid information using AIS message type 21, the Aids-to-Navigation Report.[2] U.S. AtoNs are identified by MMSI numbers starting with 993 and listed in the United States Coast Guard (USCG) Light List [26]. Aids to navigation include lights, sound signals, buoys, daybeacons, and other navigational aids.

2.1.5. AIS Search and Rescue Transmitter (SART)

AIS-SARTs aid in locating lifeboats and life rafts by transmitting safety-related text using AIS message type 14 to indicate the relevant SART status, such as active or test mode. When active, an AIS-SART also transmits position reports using AIS message type 1 with navigational status 14, using AIS-SART burst transmissions [29]. AIS-SARTs are also used in maritime survivor locating devices (MSLDs) or man-overboard (MOB) devices. Standard AIS-SARTs can be identified by MMSI numbers beginning with 970 [30].

2.1.6. Search and Rescue Aircraft

Search and rescue aircraft report position every ten seconds [25] using message type 9.[3]

2.2. Security-Relevant AIS Use

AIS supports collision avoidance, vessel traffic services, maritime-domain awareness, AtoN reporting, search and rescue, accident investigation, and the broadcast of navigational or environmental information, including application-specific messages (ASMs) [24,25]. These functions include both physical and virtual navigation aids, such as virtual markers or warnings about restricted areas. In the security context, the important point is that AIS messages are consumed by bridge displays, shore monitoring systems, satellite receivers, and public or commercial data feeds. Consequently, falsified, delayed, suppressed, or replayed AIS messages can affect both local navigation decisions and wider maritime-domain awareness.

2.3. How AIS Works

2.3.1. TDMA Protocol

AIS uses TDMA-based access over two VHF channels. Class A stations mainly use Self-Organized Time-Division Multiple Access (SOTDMA), Class B stations commonly use Carrier-Sense TDMA (CSTDMA) or SOTDMA variants, and base stations or AtoNs may use fixed, assigned, or random-access mechanisms such as FATDMA and RATDMA [23,24]. This shared-channel design is central to AIS operation and also to its security exposure: AIS depends on common timing, a fixed number of transmission slots in each one-minute frame, slot reservation or assignment, and channel-management messages. These mechanisms can be abused in attacks such as slot starvation, malicious channel switching, and timing disruption, as discussed in Section 3.3.6. Figures 1–3 summarize the slot structure.

The link layer controls access to the VHF channels using TDMA techniques and defines frame, slot, and message formats. Each AIS frame lasts one minute, starts at a UTC-synchronized minute boundary, and is divided into 2250 transmission slots per channel. Since AIS operates on two VHF channels in parallel; there are 4500 slots per minute across the two channels. Each slot is approximately 26.67 ms long and carries a 256-bit burst/frame at 9.6 kb/s. This 256-bit quantity is the complete slot frame, not the AIS payload alone: it includes transmitter ramp-up, preamble/training sequence, start and end flags, the AIS payload field, the frame check sequence (FCS), and the guard interval. The actual AIS payload is smaller and depends on the message type; for example, a single-slot Class A position report has a 168-bit payload, including the message ID, MMSI/user ID, navigation data, and communication-state field. Figure 3 illustrates this slot structure and an example payload breakdown.

AIS data originates from vessel navigation sensors, including a GNSS receiver and a gyroscopic compass. Static information like the vessel's name and VHF call sign is programmed into the AIS system during installation and transmitted regularly.

AIS information is displayed on dedicated equipment or integrated into electronic navigational charts, providing crew situational awareness and aiding collision avoidance.

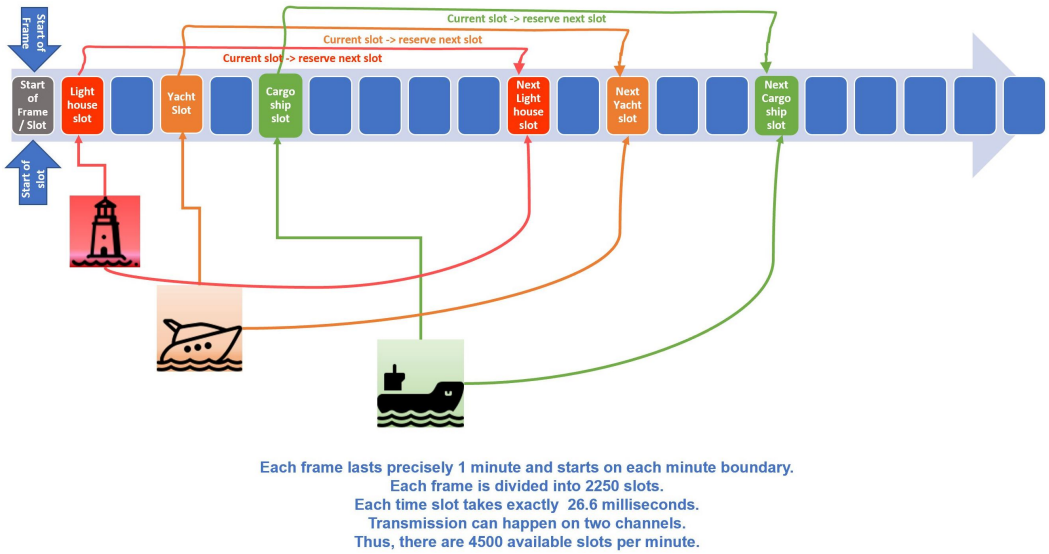


Figure 1. AIS slot communication scheme [31]. Each UTC-synchronized minute is divided into 2250 time slots (≈26.67 ms). Stations access slots by reservation (SOTDMA, Class A), carrier-sense use of free slots (CSTDMA, Class B), or fixed assignments (FATDMA for AtoN/base). Two VHF channels operate in parallel, each with the same frame structure.

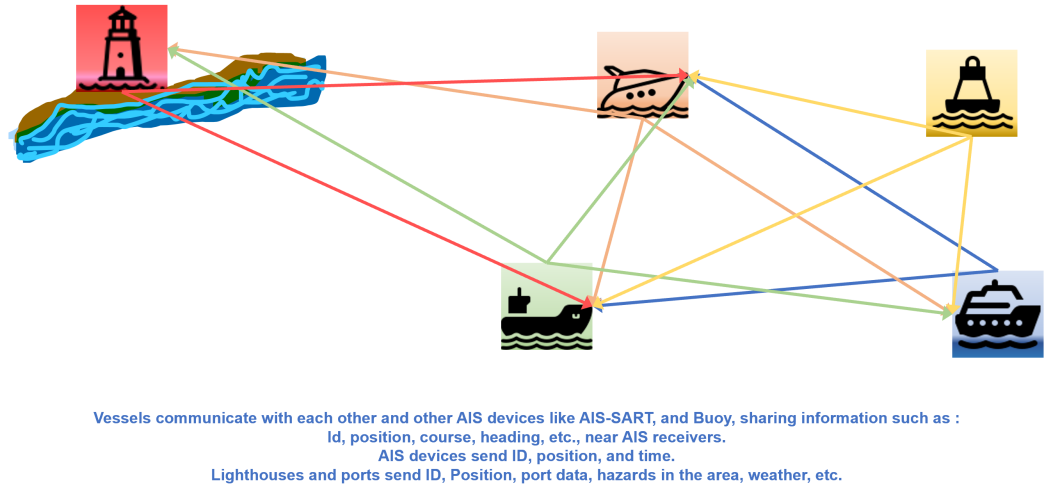


Figure 2. AIS broadcast communication scheme [31]. AIS messages are one-to-many broadcasts received by all stations within VHF range, including ships, AtoNs, buoys, lighthouses, and coastal receivers. Lines illustrate simultaneous reception; both AIS channels carry such broadcasts in parallel.

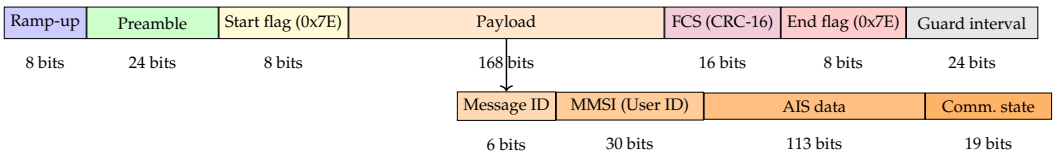


Figure 3. AIS single-slot burst (one VHF channel). At 9.6 kb/s each TDMA slot carries a 256-bit frame: 8-bit TX ramp-up, 24-bit preamble, 8-bit start flag (0x7E), payload (example shown: 168-bit Class A position report), 16-bit FCS (CRC-16), 8-bit end flag (0x7E), and a 24-bit guard interval. Example payload fields: Message ID (6 b), MMSI/User ID (30 b), AIS data (113 b), and communication state (19 b; SOTDMA). Other message types may differ in length, and some span multiple slots.

2.3.2. Security Implications of TDMA

TDMA enables many AIS stations to share the same VHF data link by dividing each one-minute frame into a fixed number of transmission slots [23,24]. This improves channel sharing under normal operation, but it also creates security-relevant dependencies on

common timing, slot reservation or assignment, and channel-management messages. In dense traffic or under deliberate interference, slot contention and channel congestion can delay or suppress safety-relevant messages. Attackers may exploit these mechanisms to create denial-of-service conditions, starve legitimate transmissions, force malicious channel switching, or disrupt situational awareness [7,32]. Thus, the TDMA design is not only an efficiency mechanism but also part of the AIS attack surface.

2.4. AIS Messages

AIS messages are defined by the AIS protocol, while NMEA 0183 specifies a standard interface format used to exchange AIS data between maritime devices and software systems [23,26,33]. The AIS message identifier is a 6-bit field, allowing 64 possible values; types 1–27 are currently defined, as listed in Table 1, and the remaining values are reserved.

Table 1. List of AIS Message Types.

Msg. ID	Name	Description	Priority	Typical Access Scheme
1	Position Report Class A	Scheduled position report from Class A shipborne equipment; provides vessel position, speed, and course	High	SOTDMA
2	Position Report Class A (Assigned)	Assigned scheduled position report from Class A shipborne equipment; provides vessel position, speed, and course	High	SOTDMA
3	Special Position Report (Response to Interrogation)	Position report transmitted in response to an interrogation	High	SOTDMA or RATDMA
4	Base Station Report	Sent by base stations with time and location information	High	FATDMA
5	Static and Voyage Related Data	Ship's static details and voyage information	Medium	SOTDMA
6	Binary Addressed Message	Binary data addressed to specific targets	Low	FATDMA or RATDMA
7	Binary Acknowledge	Acknowledgment of receipt of a binary message	Low	FATDMA or RATDMA
8	Binary Broadcast Message	Broadcast binary data transmission	Low	FATDMA or RATDMA
9	Standard SAR Aircraft Position Report	Position report from search and rescue aircraft	High	RATDMA
10	UTC/Date Inquiry	Request for UTC/date information from a station	Low	FATDMA or RATDMA
11	UTC/Date Response	Response with UTC/date information	Low	FATDMA or RATDMA
12	Addressed Safety-Related Message	Safety-related text to a specific station	Medium	FATDMA or RATDMA

Table 1. *Cont.*

Msg. ID	Name	Description	Priority	Typical Access Scheme
13	Safety-Related Acknowledgment	Acknowledgment of addressed safety-related message	Medium	FATDMA or RATDMA
14	Safety-Related Broadcast Message	Broadcast safety-related text message	Medium	FATDMA or RATDMA
15	Interrogation	Request for specific information from another AIS station	Low	FATDMA or RATDMA
16	Assignment Mode Command	Assigns specific time slots to a station	Low	FATDMA or RATDMA
17	GNSS Broadcast Binary Message	Broadcast of GNSS corrections for navigation accuracy	High	FATDMA
18	Standard Class B Equipment Position Report	Position data for Class B vessels	Medium	CSTDMA or SOTDMA (Class B SO)
19	Extended Class B Equipment Position Report	Additional voyage/static data for Class B vessels	Medium	CSTDMA or SOTDMA (Class B SO)
20	Data Link Management Message	Management of data link parameters	Low	FATDMA
21	Aids-to-Navigation Report	Position/status of an aid to navigation (e.g., buoy, beacon)	High	FATDMA or RATDMA
22	Channel Management	Management of AIS channels	Low	FATDMA
23	Group Assignment Command	Command assigning actions to a group of AIS stations	Low	FATDMA or RATDMA
24	Static Data Report (Class B)	Static data such as name and call sign from Class B equipment	Medium	CSTDMA or SOTDMA (Class B SO)
25	Single Slot Binary Message	Single-slot binary message for addressed or broadcast communication	Low	FATDMA or RATDMA
26	Multiple Slot Binary Message with Communications State	Multi-slot binary message with communication-state information	Low	FATDMA or RATDMA
27	Long-Range AIS Broadcast Message	Position report for long-range applications (reduced content/rate)	High	RATDMA

3. Threats and Vulnerabilities in AIS

This section surveys the evolving landscape of AIS threats, focusing on protocol-, signal-, and data-layer vulnerabilities and known exploitation techniques. We highlight how AIS's open, self-reported broadcast model and dependence on GNSS expose it to spoofing, jamming/denial-of-service, and data manipulation attacks. We treat mitigation strategies in Section 4, where we group defenses into four families: protocol/authentication, endpoint/network hardening, analytics for detection, and resilience/governance.

3.1. Relation to Prior Surveys

Prior surveys cover several complementary strands: AIS anomaly detection, vessel-behavior analysis, and trajectory prediction [34–36]; maritime data mining, AIS applications, AI/big-data methods, and general anomaly-detection techniques [37–41]; broad maritime cybersecurity, incidents, policy, human and organizational factors, MASS, and supply-chain security [42–49]; and supporting capabilities such as multi-sensor fusion, vision-based

maritime awareness, and maritime MLOps [50–52]. Supplementary Table S1 provides a structured overview of these surveys and their relevance to AIS security.

Compared with these works, our review is AIS-first and security-focused. Rather than treating AIS mainly as a data source, application platform, or one component of broader maritime cybersecurity, we start from AIS protocol design and operation and link AIS weaknesses to attack classes, operational impacts, and mitigation families. This yields a unified view that connects protocol-level weaknesses, attack techniques, detection approaches, resilience mechanisms, and governance measures across the AIS ecosystem.

This threat framing is also consistent with the operational perspective of IALA Guideline G1181 on VHF Data Exchange System (VDES) VHF Data Link (VDL) integrity monitoring [53]. Although G1181 addresses VDES broadly, including AIS, Application-Specific Messages (ASMs), and VHF Data Exchange (VDE), its vulnerability sources align with the AIS threat classes considered here: unauthorized signaling, misbehaving or incorrectly configured devices, unauthorized messages, spoofing, GNSS/DGNSS spoofing and jamming, denial-of-service, protocol attacks, and radio interference. Thus, G1181 provides an operational integrity-monitoring view that complements our AIS-focused cybersecurity survey.

3.2. The Growing Threat Landscape

The recent literature on maritime digitalization, Industry 4.0, IoT/cloud systems, and IT/OT convergence describes the broader connected environment in which AIS data are produced, consumed, redistributed, and fused with other ship, shore, port, and logistics systems [54–58]. In this environment, AIS-specific risk arises from AIS's own design and deployment properties: open VHF broadcast; self-reported data; lack of built-in authentication, integrity, and replay protection; GNSS dependence; shared TDMA-based access; and widespread consumption by bridge displays, shore monitoring systems, satellite receivers, and public or commercial data feeds.

Figure 4 summarizes this AIS-focused threat landscape by connecting contextual drivers, AIS attack vectors, immediate technical effects, and downstream operational and safety impacts. The remainder of this section follows this logic: we first discuss AIS-specific vulnerabilities and exploitation methods, and we then summarize their operational consequences.

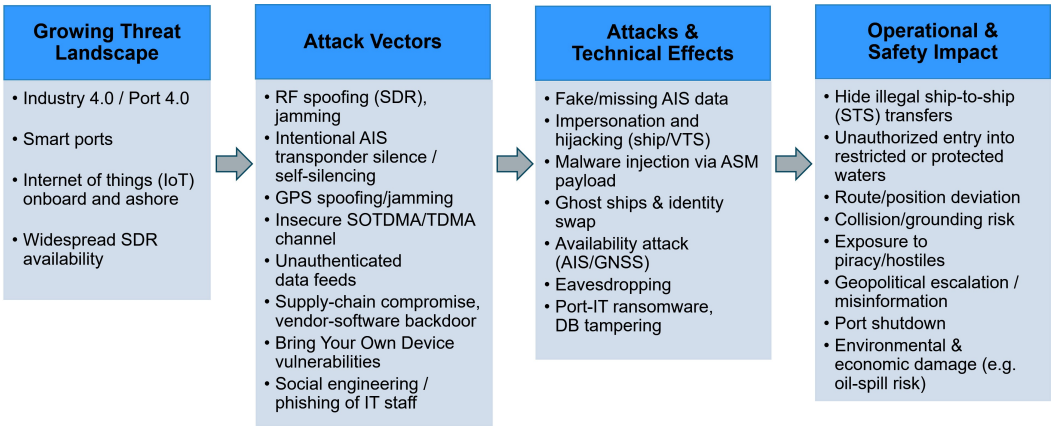


Figure 4. AIS threats flow: contextual drivers, attack vectors, technical effects, and operational/safety impacts.

Common risk factors in the maritime environment include legacy equipment, heterogeneous vendors, limited onboard cybersecurity expertise, uneven training, weakly standardized security practices, and increasing dependence on connected navigation technologies such as AIS, ECDIS, ARPA, and GNSS [57,59–61]. For AIS, these contextual factors matter because AIS messages are openly broadcast, self-reported, and used by bridge systems, shore monitoring systems, satellite receivers, and public or commercial data services.

Consequently, falsified, delayed, suppressed, or replayed AIS messages can affect both local navigation decisions and wider maritime-domain awareness.

Prior work also shows that technical defenses must be combined with governance, training, compliance monitoring, and collaboration. Studies of maritime cyber readiness, naval cyber operations, training, system preparedness, and policy gaps point to the need for anomaly detection, cross-sensor validation, structured training, incident response, and continuous adaptation [62–65]. The next subsection focuses on AIS-specific vulnerabilities and exploitation methods.

3.3. AIS Vulnerabilities and Exploitation

Developed in the late 1990s, the Automatic Identification System (AIS) underpins modern maritime safety, traffic management, and collision avoidance. It frequently broadcasts dynamic data such as position, course, and speed, and it periodically broadcasts identity, static, and voyage-related details such as MMSI, ship name, vessel type, and cargo-related information, over a Self-Organized Time-Division Multiple Access (SOTDMA) VHF link [23,24]. Because AIS relies on unencrypted, unauthenticated self-reported data and lacks message-integrity checks and replay protection, that is, built-in mechanisms for detecting tampered, duplicated, or stale AIS messages, it is vulnerable to spoofing, eavesdropping, and message manipulation [66,67]. These gaps can disrupt traffic flow, compromise vessel safety, and facilitate illicit activities ranging from smuggling and piracy to illegal fishing.

To avoid mixing design weaknesses, attacks, technical effects, and operational consequences, the discussion below is aligned with the threat-flow taxonomy in Figure 4. We distinguish (i) contextual drivers and enabling AIS weaknesses, such as open broadcast, self-reported data, GNSS dependence, shared TDMA-based access, and lack of built-in authentication; (ii) attack vectors and exploitation mechanisms, such as spoofing, jamming, channel-management abuse, and feed-side manipulation; and (iii) technical and operational impacts, such as false or missing tracks, degraded AIS availability, misleading bridge displays, and impaired traffic monitoring. The subsections below use this distinction as an organizing guide, while representative examples and case studies illustrate how the corresponding attack mechanisms produce technical and operational effects.

A substantial body of research addresses these protocol and ecosystem weaknesses. In line with the VDL-integrity concerns identified in IALA Guideline G1181 [53], several recurring AIS-specific weakness and attack categories appear throughout the literature [7,32,68]. Data falsification and spoofing attacks alter or fabricate vessel, AtoN, SART, weather, or other AIS messages [2,32]. GNSS-related attacks manipulate the position or timing inputs on which AIS reports depend. AIS VHF-link availability attacks, including jamming, flooding, slot starvation, and timing disruption, degrade or suppress legitimate AIS communication [7,68]. Signaling and channel-management attacks abuse AIS management mechanisms to alter slot use, channels, or reporting behavior. Implementation and configuration weaknesses arise from misconfigured, non-compliant, or vulnerable devices and software. Finally, feed- and shore-side manipulation affects public, commercial, or authority AIS data streams even when the local RF AIS picture is unchanged [6,69]. The following subsections use these categories to organize representative weaknesses, attacks, manipulation techniques, and case studies.

3.3.1. Data Falsification and Spoofing Attacks

Malicious actors can manipulate AIS data, for example, by altering location information to conceal illicit activities, including piracy, smuggling, and unauthorized fishing in restricted zones. Although AIS provides valuable opportunities for monitoring fishing

activities, it also presents significant challenges due to its reliance on self-reported data, which is vulnerable to falsification [70,71]. Deliberate AIS signal disruptions, particularly "silence anomalies," where vessels disable their AIS transponders to engage in illegal fishing, are commonly observed near Exclusive Economic Zones (EEZs) and Marine Protected Areas (MPAs) [72].

3.3.2. TDMA/VDL Access Weaknesses

The TDMA protocol underpinning AIS lacks essential security features such as encryption and authentication, exposing it to spoofing, data corruption, and hijacking [23,24].

3.3.3. Eavesdropping and Unauthorized Reception

The unencrypted nature of AIS transmissions allows attackers to intercept sensitive information, increasing the risk of unauthorized access [73].

3.3.4. AIS Feed and Data-Source Manipulation

Many public ship-tracking websites rely on unauthenticated data feeds, making them vulnerable to manipulation through fake AIS packets transmitted via Software-Defined Radio (SDR). Androjna and Perkovič [4] conducted a case study examining several vessels involved in AIS data manipulation. This includes recent incidents related to Ship-to-Ship (STS) transfers, a common practice in the shipping industry for transferring large quantities of oil or other cargo between vessels. Their investigation into suspicious STS oil transfers revealed falsified or missing position data and confirmed that AIS messages can be spoofed, disrupted, deliberately falsified, or simply switched off.

3.3.5. AIS VHF-Link Availability Attacks: Jamming, Flooding, and Disruption

AIS VHF-link availability attacks target the ability of legitimate AIS messages to be transmitted, received, or processed. Examples include VHF jamming and interference, SDR-based flooding, slot starvation, timing disruption, and abuse of channel-management mechanisms [7,32,68]. These attacks can delay, suppress, or overload AIS communication and thereby degrade situational awareness.

3.3.6. Representative AIS Manipulation Techniques and Case Studies

While AIS spoofing involves transmitting false data to misrepresent a vessel's identity or location, requiring control over an AIS transmitter, AIS jamming disrupts communication by overwhelming the system with high-power transmissions, preventing data reception without sending false information [7].

AIS position reports and TDMA timing depend on GNSS-derived inputs, making AIS operation sensitive to GNSS jamming and spoofing. GNSS attacks do not necessarily spoof AIS messages directly; rather, they can corrupt the position, speed, course, or timing information that AIS equipment uses and broadcasts. When combined with AIS's lack of built-in authentication, integrity protection, and replay protection, such manipulation can produce false, missing, delayed, or implausible vessel tracks. These effects may be exploited to conceal illicit activity, create misleading bridge or shore displays, or intensify uncertainty during maritime incidents and territorial disputes [2,3,74,75].

While Class B AIS transponders exhibit some resilience to spoofing, owing to factors such as type-approval testing[4], reliance on an internal GNSS receiver, often GPS, and single-slot transmission (which limits the complexity of spoofed messages compared to the multi-slot transmissions used by larger vessels), they nonetheless remain vulnerable to sophisticated forms of manipulation.

AIS spoofing has been used in various contexts, ranging from electronic warfare (EW) to conceal military maneuvers, to non-military applications intended to disrupt

commercial shipping operations [5,69]. Zorri et al. [76] provide a detailed analysis of AIS spoofing incidents, including cases involving NATO vessels in the Black Sea and commercial ships in the Strait of Hormuz, highlighting the absence of authentication and data integrity mechanisms in the AIS protocol. Their study emphasizes the urgent need for cryptographic safeguards and the adoption of resilient, terrestrial navigation backups such as eLORAN (Enhanced Long Range Navigation) [77], a ground-based radio navigation system designed to provide robust positioning, navigation, and timing (PNT) services even in the event of satellite system disruption or spoofing. Additionally, the research underscores the importance of training maritime personnel to detect AIS anomalies and advocates for implementing multi-layered security architectures to strengthen maritime situational awareness and operational resilience.

Grant et al. [78] further highlight the critical impact of GPS jamming on AIS functionality, demonstrating how signal disruption leads to erroneous AIS positions, compromised data synchronization, and significant maritime hazards. The authors advocate for integrating eLORAN as a backup system for PNT to mitigate these vulnerabilities, stressing the importance of redundant navigation systems and comprehensive crew training in alternative navigation methods to ensure operational continuity in the face of GPS denial attacks.

Androjna et al. [2,3,75,79] analyze AIS/GNSS vulnerabilities and their effects on maritime situational awareness. Their case studies, including the Elba Island spoofing incident, show how false AIS or GNSS-derived information can create misleading vessel positions, disrupt navigation displays, and increase the load on maritime monitoring systems. For the purposes of this survey, the AIS-specific lesson is that unauthenticated AIS data, GNSS dependence, and SDR-enabled injection can combine to produce false tracks, missing or implausible positions, and misleading operational displays.

Balduzzi et al. [32] conducted a comprehensive security evaluation of AIS, highlighting critical vulnerabilities in both the protocol's specifications and its software and radio frequency (RF) implementations. Their study revealed that AIS is susceptible to spoofing, hijacking, and denial-of-service attacks, which can undermine vessel navigation, collision avoidance, and maritime traffic management. The authors developed an AIS VHF Data Message (AIVDM) encoder to generate and inject arbitrary AIS messages, enabling the creation of false vessel signals, spoofed navigational aids, and bogus distress alerts. Additionally, they implemented AISTX, a software-defined radio-based AIS transmitter, to demonstrate real-world signal manipulation over distances of up to 16.5 km. Their attack simulations exposed vulnerabilities such as slot starvation, frequency hopping, and AIS hijacking, all of which pose serious threats to the integrity and availability of AIS communications.

Spoofing Techniques and Examples: Spoofing methods include using AIS base stations, SDR platforms, and transponder interface exploitation to transmit unauthorized messages [7,78]. Several techniques have been used to demonstrate and conduct AIS spoofing attacks, with varying degrees of sophistication and impact.

Vessel spoofing: In this technique, an attacker broadcasts falsified AIS information to create a phantom vessel: static fields such as MMSI or call-sign and, more critically, dynamic fields like position, speed, or course. By manipulating those dynamic parameters, the adversary can mask illicit behavior, including unauthorised approaches, incursions into restricted zones, route deviations, or suspicious AIS on/off cycles [32]. DiRenzo et al. [80] highlight AIS exposure to spoofing, jamming, and denial-of-service, and advocate cryptographic authentication together with eLORAN as a complementary navigation system. Mednikarov et al. [81] reach similar conclusions, stressing network segmentation, intrusion detection, and encryption to bolster AIS security.

GNSS Spoofing: Androjna et al. present a comprehensive analysis of spoofing incidents targeting Global Navigation Satellite Systems (GNSS) in maritime traffic between 2008 and 2020, highlighting the growing threat to systems reliant on satellite-based positioning [2,3,74].

Territorial disputes, heightened tensions, and AIS data-layer injection: Harris [6] documents incidents in which false AIS tracks for naval vessels appeared in public AIS aggregators' feeds, with the available evidence pointing to feed-side or aggregator-level manipulation rather than confirmed over-the-air VHF spoofing. Such cases show that adversaries can remotely place phantom ships on public AIS maps without access to shipboard equipment. However, because these manipulations do not necessarily alter the local RF AIS picture used by nearby ships and authorities, their primary impact is to sow confusion, create false narratives, and heighten tensions in contested waters.

The Crimea/Sevastopol incident illustrates this risk. In June 2021, public AIS data showed HMS Defender and HNLMS Evertsen leaving Odessa and approaching Sevastopol, although contemporaneous open-source checks, including webcam and witness evidence, indicated that both vessels were still docked in Odessa [5,6]. The scale of this incident was therefore narrow, two naval vessels and a false transit track, but the geopolitical sensitivity was high. Similarly, bogus Swedish-navy tracks near Kaliningrad in March 2021 reportedly involved nine Swedish naval vessels and were publicly denied by the Swedish Navy [6,69]. A related sanctions-evasion example involves North Korean vessels cycling or altering AIS identities to obscure activity, which reflects a more persistent law-enforcement problem rather than a single bounded spoofing event [82]. Across these examples, the evidence is strongest for the appearance of false tracks or identity manipulation in public AIS data. However, the available sources do not conclusively establish who injected the data, how it entered the AIS data stream, or whether the false tracks were also received over the local VHF AIS channel and displayed on operational bridge systems.

Real-World Test and Academic Research: A 2013 University of Texas at Austin experiment successfully spoofed a yacht's GPS, exposing vulnerabilities in onboard navigation systems [44,83]. Similarly, a 2008 UK trial using GPS jamming equipment demonstrated potential disruptions to onboard AIS and navigation systems [78,84].

Fake Weather Forecasts: One application of AIS is the communication of dynamic data reflecting the changing environment, such as currents and climate conditions. A special type of message, namely a binary message, is used to convey such information. Spoofing may be used to broadcast false weather forecasts, for example, reporting heavy fog when visibility is actually clear [32].

Navigation and Closest Point of Approach (CPA) Spoofing: AIS reports can be used by bridge systems, such as ECDIS, radar/ARPA, or collision-avoidance displays, to calculate the Closest Point of Approach (CPA) between vessels based on reported positions and trajectories. Such systems may generate visual or audible alerts when a potential collision is detected. Deliberately fabricating a false imminent-collision scenario can therefore trigger misleading CPA alerts, potentially prompting the officer of the watch (OOW) to consider unnecessary or hazardous evasive action. In constrained waters, such responses could increase the risk of collision, grounding, or contact with hazards such as rocks or shoals.

The Elba Island incident illustrates this type of display-level risk. In December 2019, a large but short-lived AIS spoofing event occurred near Elba Island, generating phantom AIS targets and disrupting the local AIS picture [2,3]. The reported evidence comes from AIS data later replayed and analyzed by Androjna et al. [2], rather than from a contemporaneous on-board incident report. Their analysis identified thousands of generated AIS messages and hundreds of false tracks in the Elba–Corsica area. Using the reconstructed

data, they simulated the information that would have appeared on the ECDIS screen of one of the ships present during the incident. The reconstructed bridge display, with radar/ECDIS and AIS overlay, would have shown a hazardous situation in which the vessel appeared to be on a collision course with over a dozen phantom merchant vessels, potentially prompting inappropriate OOW decisions. Thus, the case is best understood as a reconstructed bridge/VTS-display effect: it illustrates how spoofed AIS data could create misleading collision cues and increase processing load, but the available sources do not establish the attacker, motive, or an actual collision or casualty.

AIS-SART Spoofing: As pointed out by Androjna and Perkovič [74], False Search and Rescue Transponder (SART) alerts can lure vessels into hostile waters under the pretense of aiding distressed ships.

Denial-of-Service (DoS) Attacks: Denial-of-Service (DoS) attacks on AIS aim to disrupt maritime communication by overwhelming or manipulating the system, thereby hindering the transmission and reception of critical navigational data. Several studies have examined the impact of DoS attacks on AIS communication [7,32,80]. Khandker et al. [7] transmitted 200,000 AIS type 1 frames using an SDR. Their evaluation showed that nearly 90% of the tested hardware/software configurations were affected by the DoS attack, resulting in issues such as output congestion, unresponsiveness, or system crashes. Balduzzi et al. [32] identified and analyzed several radio-frequency-based DoS techniques targeting AIS communication:

1. **Slot Starvation:** This technique prevents legitimate AIS communication by exploiting transmission slot allocation mechanisms. It can be executed in two primary ways:
    (a) **Impersonation:** Attackers impersonate maritime authorities to reserve the entire AIS transmission "address space," effectively occupying all available slots. This action prevents other AIS stations within range from transmitting, leading to widespread communication failure.
    (b) **Flooding:** Attackers send a large volume of fabricated AIS messages, consuming available bandwidth and processing capacity. As a result, legitimate messages may be delayed, dropped, or ignored, impairing vessel coordination and situational awareness.
2. **Frequency Hopping:** Attackers impersonate authoritative entities to issue commands that force AIS transponders to switch frequencies. This tactic disrupts communication and undermines system reliability.
3. **Timing Attacks:** These attacks exploit AIS timing mechanisms by repeatedly altering transmission schedules, delaying messages, or sending renewal commands. The resulting interference may cause vessels to disappear from ECDIS displays, leading to navigational confusion and increased collision risk.

Aids to Navigation (AtoN) Spoofing: Aids to Navigation (AtoNs), such as buoys and lighthouses, are critical for managing vessel traffic and warning of navigational hazards like shallow waters and rocks. AtoN spoofing refers to generating false or misleading information about these aids, potentially deceiving vessels, and compromising navigational safety. For example, attackers may fabricate nonexistent buoys near a harbor entrance to disrupt maritime traffic or mislead vessels into hazardous areas, increasing the risk of collisions, groundings, or other incidents. Balduzzi et al. [32] demonstrated that their SDR-based attack successfully injected fake AtoNs. As noted by Androjna et al. [2], in a 2020 incident at Ponce de Leon Inlet, Florida, attackers used AIS spoofing to inject entirely fabricated virtual buoys, illustrating how digital AtoNs can be manipulated to threaten maritime safety.

AIS Hijacking: As defined by Balduzzi et al. [32], AIS hijacking refers to the manipulation or overriding of the AIS signal of a real, existing vessel. In such an attack, the adversary attempts to assume a legitimate vessel's identity or communication channel and alters its broadcasted data (such as location, speed, heading, or identity). Walde and Einar [85] demonstrate that AIS hijacking is feasible using AISTX, an open-source software-defined AIS transmitter developed by Balduzzi et al. [32].

Application-Specific Messages (ASMs): Although not directly involved in manipulating core AIS data, sophisticated attackers can exploit binary messages to inject malware and compromise vessel systems. Amro et al. [86] explored cybersecurity threats in maritime environments by demonstrating how AIS can serve as a covert command-and-control (C&C) channel, namely a communication channel through which an attacker can issue instructions to compromised shipboard systems. Their work highlights inherent AIS vulnerabilities, particularly its lack of encryption and authentication, which make it susceptible to unauthorized transmissions and cyber manipulation.

The authors introduced a novel attack vector in which AIS messages are used to transmit commands to compromised shipboard systems, underscoring the risks to autonomous maritime operations. Using the MITRE ATT&CK framework, a knowledge base of adversary tactics and techniques [87], they developed a threat model focusing on command-and-control (C&C) and defense-evasion techniques. Their proof of concept demonstrated how encrypted commands could be embedded within AIS Type 8 messages using AES encryption and Base32 encoding. Furthermore, Amro et al. [86] simulated two cyber kill chains, that is, step-by-step attack sequences from initial compromise through command-and-control and follow-on actions, targeting an autonomous passenger ship (APS), illustrating the feasibility of remote malware control via AIS.

3.3.7. Operational Consequences of AIS Manipulation

AIS manipulation can have far-reaching consequences across various domains:

Safety and Navigation: Manipulated AIS data can mislead vessels, potentially causing deviations from planned routes, collisions, groundings, and other navigational hazards. Such incidents endanger crew safety and pose significant risks of environmental damage, including oil spills and marine ecosystem disruptions.

Khandker et al. [7] illustrate the operational impact of AIS availability attacks. In their SDR-based experiments, crafted AIS message floods affected nearly 90% of the tested hardware/software configurations, causing output congestion, unresponsiveness, or crashes. These results show that AIS spoofing, jamming, and denial-of-service attacks can degrade vessel tracking, collision avoidance, and maritime traffic-management functions.

Security and Law Enforcement: The transmission of falsified AIS information can obscure a vessel's identity, route, or activity, hindering law-enforcement efforts to monitor illicit behavior such as sanctions evasion, suspicious ship-to-ship transfers, illegal fishing, smuggling, or piracy [4,72,82,88]. This compromises maritime security and can facilitate unlawful operations.

Geopolitical and Strategic: Fabricating AIS data in sensitive maritime regions can heighten geopolitical tensions by creating false narratives about vessel presence, movement, or intent. Reported cases involving forged naval AIS tracks near contested waters illustrate how AIS manipulation can be used in information operations and may contribute to uncertainty, miscalculation, or escalation [5,6,69,76].

4. Proposed AIS Security Measures

Although AIS plays a crucial role in maritime safety, navigation, and traffic management, it remains susceptible to cyber threats such as spoofing, hijacking, data manipulation,

and denial-of-service, posing substantial risks to maritime security and operational reliability [88]. This motivates the deployment of effective security measures.

This section surveys measures proposed in the literature and organizes them along two axes: (i) categories: Protocol and Cryptography, Endpoints and Network, Analytics and Detection, Resilience and Redundancy, and Governance, People, and Ecosystem; and (ii) incident lifecycle phases: Prevent (P), Detect (D), Respond (R), and Recover (Rv). The taxonomy appears in Figure 5. The measures are designed to operate within AIS's open architecture while accommodating key management constraints and maintaining backward compatibility with legacy maritime systems.

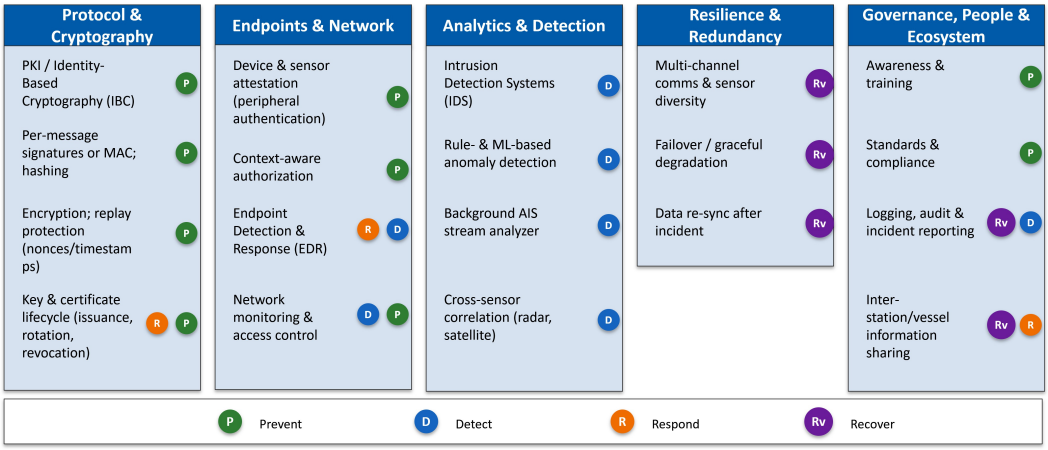


Figure 5. AIS Security Measures: Categories and Lifecycle Coverage.

4.1. Design Constraints for AIS Security Measures

4.1.1. Open Broadcast Model

AIS relies on an open Self-Organized Time-Division Multiple Access (SOTDMA) broadcast without built-in authentication, integrity, or replay protection. These properties enable spoofing and data manipulation. Any enhancement must preserve interoperability and receiver compatibility across the installed base.

4.1.2. Key Management

Cryptographic assurances, such as digital signatures, message authentication codes (MACs), encryption, and replay protection, depend on scalable key management across a worldwide fleet with intermittent connectivity and diverse equipment. Approaches include public key infrastructure (PKI) and context-bound symmetric authorization (e.g., deriving short-lived keys from time and small geographic tiles using a lightweight cipher such as RC5 [89], suitable for underwater assets that surface only briefly) [90]. Keys should be generated under audited multi-person control and distributed securely when assets surface; lifecycle operations (issuance, rotation, revocation/status) must be feasible at sea.

4.1.3. Backward Compatibility

Security enhancements must coexist with legacy AIS transponders and infrastructure. Compatibility can be maintained by deploying authentication as an overlay rather than by changing baseline AIS message semantics [67,91]. In such a staged design, ordinary AIS position, static, and voyage messages continue to be transmitted and processed normally, while upgraded ship or shore systems exchange additional authentication material, such as certificates, signatures, or MACs, using standard-compliant optional or application-specific messages, or over ASM/VDES channels where available [92,93]. The extension should be designed so that non-upgraded receivers are not exposed to non-standard message

formats and continue to process ordinary AIS traffic normally; interoperability and type-approval testing are therefore essential. Upgraded receivers can verify the additional authentication material and mark AIS targets as authenticated, unauthenticated, or low-confidence. During partial deployment, unauthenticated AIS messages should not be discarded by default; instead, they should remain visible but be treated as unverified and cross-checked against radar, electro-optical/infrared (EO/IR), GNSS consistency, or shore-side validation [94,95].

4.2. Security Measures by Category

4.2.1. Protocol and Cryptography

Per-message authentication and confidentiality. Message-authentication codes (cryptographic MACs), digital signatures such as the Elliptic Curve Digital Signature Algorithm (ECDSA), authenticated encryption with associated data (AEAD), replay-protection mechanisms based on timestamps or nonces, i.e., values used once to detect stale or duplicated messages, and even signal-domain watermarking have been proposed to provide authenticity and integrity for AIS payloads [34,66,93]. Public-key cryptography [96] has likewise been explored for AIS messages (including Message 8, see Table 1) [67,93]. Because baseline AIS lacks these primitives, effective key management is a prerequisite for any deployment [97]. Taken together, these controls prevent spoofing and in-transit manipulation by binding each message to a verifiable sender and rejecting replays, and support response through rapid credential revocation and re-keying after compromise.

Identity-based and modular authentication. SecAIS leverages Identity-Based Cryptography (IBC) with symmetric anonymization to simplify key distribution relative to classic PKI [66,73,98]. Auth-AIS is a modular protocol that enables flexible authentication (signatures or MACs) while maintaining backward compatibility [91,99]. These approaches primarily prevent spoofing and content tampering via authenticity/integrity. Key rollover, that is, the periodic replacement and revocation of keys, supports response to credential compromise.

Location/time-bound symmetric authorization. Time- and area-scoped keys authorize transmissions for specific contexts (e.g., underwater assets) where PKI is impractical [90]. Binding authorization to place and time prevents unauthorized transmissions outside the designated parameters.

Attribute-based security and secure aggregation. For ship- and shore-side devices, attribute-based encryption (ABE), a cryptographic approach in which access rights are tied to attributes or policies, can be combined with zero-knowledge proofs to provide policy-based, fine-grained access to data [100–102]. It also enables integrity-preserving aggregation: combining readings from many devices into summaries (such as counts or averages) with cryptographic evidence that the summary has not been tampered with, even if individual readings stay hidden. These controls prevent unauthorized access and forgery while scaling to large fleets and port installations [102].

From a deployment perspective, cryptography for AIS should distinguish authentication from confidentiality. Core safety-critical AIS broadcasts, such as position, identity, and voyage data, should remain decodable by legacy and non-participating receivers; otherwise, encryption could reduce navigational awareness. Thus, for open AIS traffic, cryptography is most feasible as an authentication, integrity, and replay-protection overlay, while confidentiality is more appropriate for auxiliary/application-specific payloads, ship–shore or shore–shore links, and VDES/ASM services with known participants [66,67,91,93]. Identity-based or modular authentication is especially applicable in staged trust domains, such as ports/VTS areas, fleets, authority-operated services, base-station/AtoN networks, SAR operations, and VDES-capable deployments, where participants, credentials, and

revocation policies can be managed. Such mechanisms are less suitable as mandatory global AIS requirements until broad coordination, credential distribution, revocation, and legacy compatibility are standardized.

4.2.2. Endpoints and Network

Device and sensor attestation and RFF Peripheral authentication and context-aware authorization act preventively by admitting only devices that can cryptographically attest their identity and state (e.g., challenge–response with signed measurements), blocking rogue or tampered sensors; they act detectively by continuously checking attestation results and Radio-Frequency Fingerprints (RFFs) against enrolled profiles to flag changes in a device's RF "signature," detect copied or relayed transmitters, and highlight other signs of spoofing [103].

EDR and IDS on vessel/shore endpoints. Endpoint Detection and Response (EDR) combines continuous monitoring with automated containment or remediation on compromised endpoints. Monitoring, anomaly alerts, and forensic visibility contribute to detection, while containment and remediation actions serve the response phase. Complementing EDR, Intrusion Detection Systems (IDS) provide rule-based or anomaly-based identification of suspicious traffic, strengthening detection but typically lacking built-in response capabilities. Together, EDR and IDS extend protection beyond cryptographic mechanisms by covering runtime compromise and malicious behavior on both shipborne and shore-based AIS endpoints [104,105].

Secure transport between remote base stations using VPNs. Virtual private networks (VPNs) create encrypted, mutually authenticated tunnels between remote base stations (RBSs) and shore systems. This prevents eavesdropping, tampering, and session hijacking across these links by enforcing confidentiality and integrity at the transport/network layer, for example using IPsec or Transport Layer Security (TLS) with forward secrecy, so past sessions remain confidential even if a long-term key is later compromised. VPNs complement message-level authentication for AIS content integrity end-to-end [106].

Network monitoring and access control. Port-based network access control, such as IEEE 802.1X/EAP-TLS, forwards normal traffic only after device authentication; switchport security limits ports to known MAC addresses, disables unused ports, and shuts down ports on violations [107]. Together with least-privilege ACLs and VLAN/microsegmentation that separates IT from OT and confines breaches [108–111], these controls prevent unauthorized attachment and lateral movement, i.e., attacker spread across hosts or network segments. Flow and packet monitoring, when integrated with IDS/IPS and security information and event management (SIEM) and compared to traffic baselines, helps detect policy violations and unusual network traffic [110,111].

4.3. Analytics and Detection

4.3.1. ML/Statistical Anomaly Detection

Model-based and ML approaches, such as deep learning models [112], unsupervised clustering [113], and time-series/state-space models [114], detect motion irregularities, shutdowns, transshipment patterns, and identity/route anomalies by exploiting position/velocity/heading/rate-of-turn time series, inter-message gaps, and identity–trajectory consistency [34].

Trajectory prediction (baselines for "normal"). Sequence models [115] learn expected paths and next-step motion so that deviations can be scored as anomalies, using recurrent or other sequential architectures [116]; these strengthen detection of spoofed tracks and implausible Closest Point of Approach (CPA) alerts [117–123].

Learning normal route patterns from clustering. Clustering historical AIS trajectories learns the normal traffic context: main shipping lanes and port-approach routes, typical speed and turn-rate profiles, and usual anchorages and waiting areas for a given region, season, or vessel class [124–129]. Vessels that deviate from these learned routes or cluster structures are flagged as outliers. To keep computation tractable at scale, trajectories are first simplified (e.g., adaptive Douglas–Peucker [125]) and then processed on distributed platforms (e.g., Apache Spark [130]).

General anomaly detectors. Practical systems use simple motion features plus ML to flag vessels going dark, making unusual maneuvers, or showing long gaps/shutdowns. Typical designs combine clustering with sequence models and apply missing-data reconstruction checks; for example, they interpolate over short gaps and verify that the implied speeds/turn rates remain physically plausible [131–134].

Behavior/state classification. Supervised models classify movement or navigational status from trajectories to support rule- and risk-based alerting [135–137].

Event-type detectors. Purpose-built methods target close-approach/rendezvous and at-sea transfer patterns, improving precision for sanction-relevant behaviors [138].

Operational monitoring. Production systems turn the detection techniques described above into practical tools for watchstanders: continuous monitoring, clear and prioritized alerts (with context), and smooth integration into existing workflows for triage and follow-up [139].

4.3.2. Scalable Pipelines for Large AIS Corpora

Distributed data platforms such as Apache Spark [130] and Hadoop [140] let teams process large AIS archives in parallel, accelerating trajectory cleaning, feature extraction, and clustering for detection at scale [129,130,140]. A common pipeline first simplifies raw tracks with the Douglas–Peucker algorithm [141] to reduce points, then clusters the simplified trajectories with DBSCAN [142] to recover main routes; deviations from these routes can be flagged as anomalies [126,141,142]. Statistical models over the clustered data then estimate how typical each movement pattern is and highlight low-probability behavior, improving precision on risky patterns [143].

4.3.3. Text–Sensor Inconsistency Detection

Prieur et al. [144] propose UMBAR, an end-to-end system that ingests AIS/radar data ("hard data") and news or other open-source intelligence (OSINT; "soft data"), builds a maritime knowledge base, and cross-checks events across sources. In practice, it detects cross-source mismatches in vessel identity, location, or timing; supports response by ranking flagged items with similarity scores and short explanations for operator review; and aids recovery by preserving provenance so records can later be corrected. On its own, it does not prevent forged or replayed messages; prevention requires complementary controls such as cryptographic authentication and device attestation.

4.3.4. Background AIS Stream Analyzer

Lightweight, rule-based checks (e.g., rate limits, MMSI reuse, physically impossible movement) can detect simple abuses quickly and provide triage before ML models run [34,88].

4.3.5. CTI-Driven Maritime IoT Detection

Deep-learning pipelines that use cyber threat intelligence (CTI) together with network telemetry can classify attack types and detect new malicious patterns in maritime IoT traffic [145].

4.4. Resilience and Redundancy

4.4.1. Cross-Sensor Validation and Coverage Gaps

The fusion of AIS with independent sources such as radar, electro-optical (EO) and infrared (IR) imagery, direction finding, maritime OT data, and shore-side validation helps maintain continuity and cross-check AIS claims when AIS is incomplete or manipulated [7,35,50,51,146,147]. Satellite AIS can provide complementary reception coverage, but it remains an AIS data source rather than an independent physical sensor [148]. Empirical work also uses dedicated aerial surveys, that is, survey aircraft carrying EO/IR cameras plus an onboard AIS receiver, to estimate the share of non-AIS vessels and to correct vessel-density maps (vessels per km^2) by accounting for the surveyed area, i.e., the water surface actually observed along flight transects (transect length $\times$ effective strip width adjusted for detection probability) [149].

4.4.2. Failover, Graceful Degradation, and Data Re-Sync

In operational deployments, AIS and related systems are commonly engineered with backup channels and sensors, along with defined fallback modes to maintain service during outages or interference [50,146,147]. During the recovery phase, typical actions include reloading missing AIS data, reconciling any altered records, and preserving explicit provenance (what data came from where and when) throughout restoration, consistent with maritime cyber-resilience guidance [21,150,151].

4.5. Governance, People, and Ecosystem

4.5.1. Risk Assessment and Cyber Situational Awareness

Continuous risk assessment and cyber situational awareness (CSA) (seeing what is happening, understanding it, and anticipating what may happen next) support early warning and detection. Common tools include (i) Failure Modes and Effects Analysis (FMEA) per IEC 60812 to list and score failure modes by severity, likelihood, and detectability, and (ii) Bayesian networks (BNs) to update those risk scores as new evidence arrives (e.g., sensor readings, alerts). Many works combine the two: FMEA structures the hazards, then key factors are mapped into a BN so probabilities can be updated online [79,105,152–158]. Broader taxonomies frame vulnerabilities and mitigations at the system-of-systems scope [159].

4.5.2. Taxonomies and Holistic Frameworks

Sector-wide frameworks help plan mitigations and resilience across technical and human factors. Examples include MaCRA, which models assets, threats, and operating context (route/ship/area) to rank risks and test mitigations, and the policy framework of Tam & Jones, which maps stakeholders (ship, company, port/authority, regulator) and ship/shore attack surfaces to guide defense-in-depth and risk management [160,161].

4.5.3. Crew Training and Awareness

Training to spot GNSS/AIS inconsistencies (e.g., jumps, illogical motion, cross-sensor conflicts) and to handle systems securely prevents human-factor incidents and improves detection when data are suspect [16,63,109,162].

4.5.4. Collaboration and Information Sharing

Trusted information-sharing initiatives and verification of nearby vessel activity strengthen prevention and coordinated response [163–165].

4.5.5. Regulation and Compliance

Key frameworks and standards include: IMO/ISM—cyber risk integrated into the Safety Management System with guidance for preparation, detection, response, and restora-

tion [17,18,150,166,167]; EU NIS/NIS2—risk-management measures and incident reporting for essential/important entities [22]; Classification and national guidance—defense-in-depth objectives and operational practices [21,151,162]; and ISO/IEC 27001—an ISMS for ongoing risk treatment and improvement [168].

5. Discussion and Recommendations

This section builds on the protocol mechanics (Section 2), the threats and vulnerabilities organized as a threat taxonomy (Section 3), and the mitigation families (Section 4) to set out concise, actionable recommendations for practice and priorities for research. In brief, AIS's open, self-reported TDMA-based broadcast design, including the timing/slot structure illustrated in Figures 1 and 2, exposes known attack avenues such as spoofing, VHF-channel denial-of-service through jamming, and data manipulation. The recommendations below use the mitigation taxonomy in Figure 5 and the incident-lifecycle view of prevention, detection, response, and recovery as a basis, but they are presented as cross-cutting priorities rather than as a repetition of the taxonomy.

5.1. Protocol and Authentication Priorities

We recommend deployable authentication mechanisms that respect AIS's legacy ecosystem and compatibility constraints (Section 4) while improving message integrity and replay resistance. Practical options include carrying compact signatures or MACs in standard-compliant application-specific/binary AIS messages, or through Application-Specific Messages (ASMs) or VHF Data Exchange System (VDES) services where supported by the deployed equipment and operational setting, together with lightweight replay protection. Such mechanisms should leave baseline AIS safety messages readable by legacy receivers and allow upgraded receivers to mark targets as authenticated, unauthenticated, or low-confidence rather than suppressing unauthenticated traffic. Where end-to-end cryptographic authentication is not deployed, auxiliary physical-channel verification can provide complementary evidence linking the claimed MMSI to the observed transmitter or vessel, for example, through RF-emitter fingerprints [103] or peripheral/optical-attribute schemes adapted from vehicular systems [169–171].

Current status. No cryptographic AIS message-signing mechanism is defined in the International Telecommunication Union–Radiocommunication Sector (ITU–R) Recommendation M.1371 for operational AIS traffic; AIS management messages such as types 20, 22, and 23 support channel and group management rather than message authentication [29]. The VHF Data Exchange System (VDES) specified in ITU–R M.2092 integrates AIS, Application-Specific Messages (ASMs), and VHF Data Exchange (VDE) links, and provides higher-capacity bearers that could support authentication-related metadata or future security extensions. However, there is currently no broadly mandated IMO/ITU/IEC AIS message-signing scheme for operational AIS traffic [92]. Auxiliary physical-channel verification therefore remains an active research direction, including radio-frequency specific emitter identification (RF-SEI) and optical/peripheral attribute binding adapted from vehicular systems [172–174].

5.2. Advanced Anomaly Detection

Detection should move beyond inactivity/silence flags toward behavior-aware models (Section 4; see field semantics in Table 1 and TDMA timing in Figure 2). Models must be stress-tested against falsification tactics surveyed in Section 3 and regularly updated to track SDR-enabled attack evolution. Pointers to specific approaches appear in Section 4.3.

Current status. Behavioral anomaly/spoofing detection is widely deployed shore-side by commercial analytics providers and NGOs (e.g., "dark shipping," AIS position validation,

identity/track inconsistency detection); these are operational practices rather than prescriptive AIS standards, and no IMO/ITU/IEC text specifies particular ML models [175–178].

5.3. Fuse AIS with Independent Sensors and Cross-Check Discrepancies

In practice, AIS should be cross-checked against independent sources such as ship/shore radar, synthetic-aperture radar (SAR), electro-optical/infrared (EO/IR) imagery, direction finding, GNSS/eLORAN consistency checks, and shore-side validation. Satellite AIS can also provide complementary reception coverage, but it should be treated as an additional AIS data source rather than as an independent physical sensor. Systems should automatically flag inconsistencies between AIS claims and these sources, and they should handle flagged cases through defined incident-response procedures during the Detect and Respond phases in Figure 5. Section 4.4.1 outlines integration patterns and how to balance latency with operator triage.

Current status. Operational today on bridge and ashore: International Electrotechnical Commission (IEC) standards specify radar performance and AIS target handling/association (IEC 62388) and the portrayal of navigation information on shipborne displays (IEC 62288); IMO circulars standardize how navigation symbols are shown, and ECDIS, radar, and AIS systems implement these rules to associate AIS with radar targets and flag inconsistencies [94,95,179].

5.4. Machine Learning Operations (MLOps), Evaluation, and Datasets

Live analytics should ship with versioned datasets, leakage-safe evaluation, drift monitoring, and reproducible benchmarks tied to the misuse categories in Section 3 and the coverage taxonomy in Figure 5. This supports safe iteration as traffic patterns and attacker behavior change.

Current status. These practices are common in vendor/authority analytics stacks but are not prescribed by maritime technical standards; IMO cyber guidance is intentionally high-level and process-oriented [16].

5.5. Governance, Collaboration, and Human Factors

Because AIS operates in a heterogeneous, safety-critical ecosystem, controls must include crew training, Standard Operating Procedures (SOPs), and clear authority and hand-off rules across Vessel Traffic Service (VTS), shore-based traffic management, and bridge workflows (the ship's on-watch, pilotage, and decision-making processes), not only technical hardening (cf. Governance and Resilience in Figure 5). Where appropriate, cooperative consistency checks and secure consensus among stations (for example, Byzantine agreement to corroborate shared alerts) can strengthen collective defense [180].

Current status. At the process level, these controls are mandated via the International Safety Management (ISM) Code through IMO Resolution MSC.428(98) (requiring cyber risk to be addressed in the Safety Management System) and the updated IMO MSC-FAL.1/Circ.3/Rev.3 guidelines [16,17].

5.6. Resilience, Recovery, and Broader Implications

It is recommended to engineer for graceful degradation and rapid recovery (see Figure 5). During slot congestion or RF interference, systems should, where standards and operational control permit, prioritize safety-critical information and defer or rate-limit lower-priority data so essential collision-avoidance information continues to flow. When authentication information is unavailable or a target remains unauthenticated, displays should not simply suppress AIS targets: they should continue to show them while (i) flagging them as unverified or low-trust, (ii) throttling and inspecting suspicious streams, and (iii) validating them against other sensors such as radar, EO/IR, GNSS, or eLORAN.

Blocking every unauthenticated message can degrade situational awareness. For recovery, rely on clear, tested incident-response and restoration procedures: isolate the fault, switch to alternate sensors/links, re-synchronize and reconcile delayed or suspect data, restore normal configuration, and conduct a post-incident review.

Current status. VHF Data Link (VDL) load/channel management and group assignment are supported by AIS management messages (e.g., 20/22/23) in ITU–R M.1371 and by International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Recommendation A-124 and its appendices on channel and VDL management; VDES (ITU–R M.2092) adds extra data links so operators can prioritize traffic by service. Per IEC 62288/IMO guidance, bridge displays use standardized symbols and show lost/degraded target alerts; there is no standardized "cryptographic trust" flag because AIS has no protocol-level signing standard [29,92,94,179,181,182].

5.7. Limitations and Deployment Constraints

Even when combined, the measures surveyed above should not be interpreted as fully eliminating AIS vulnerabilities. Many weaknesses arise from AIS's fundamental design: open broadcast, self-reported data, lack of built-in authentication and replay protection, GNSS dependence, and global legacy deployment. Cryptographic protections face major deployment barriers, including global credential issuance and revocation, intermittent connectivity, type approval, interoperability with legacy receivers, and coordination across flag states, vendors, ports, VTS authorities, and commercial data providers. Confidentiality is also unsuitable for core open safety broadcasts if it hides essential traffic from non-participating receivers.

Resilience and validation measures have different constraints. eLORAN, radar, EO/IR, direction finding, satellite imagery, and shore-side validation can improve robustness, but require infrastructure, integration with bridge and shore systems, operator training, and procedures for resolving conflicting evidence. Analytics-based methods face data-quality problems, sparse validated ground truth, class imbalance, concept drift, false positives, and the need for explainable alerts that fit VTS and bridge workflows. AIS security should therefore be viewed as a staged, layered, standards-compatible improvement process rather than as a single global fix.

5.8. Priority Research Directions

The most important open research questions are deployment-oriented. First, how can AIS authentication and replay protection be deployed as standard-compliant overlays while supporting partial adoption, credential revocation at sea, and legacy interoperability [66,67,91,93]? Second, how should bridge and VTS displays represent authenticated, unauthenticated, conflicting, or low-confidence AIS targets without creating alarm fatigue or reducing situational awareness [94,95]? Third, how can public and reproducible benchmarks be built for AIS spoofing, jamming, feed manipulation, and shutdown behavior, given the scarcity of validated ground truth and the risk of overly optimistic evaluation caused by overlap between training and test trajectories [7,34,35,52]?

Additional priorities concern resilience and governance. Future work should evaluate how radar, EO/IR, SAR, direction finding, GNSS/eLORAN consistency checks, and shore-side validation can be fused under realistic latency, coverage, and cost constraints [50,51,146,147]. Finally, research is needed on governance and migration models: certification, liability, key-management authorities, standardization pathways, and incentives for adoption by vessels, ports, VTS operators, satellite/public AIS providers, and equipment vendors [16,17,109,162].

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/jmse14141258/s1: Table S1: Representative surveys relevant to AIS security, anomaly detection, and maritime cyber risk.

Author Contributions: Conceptualization, S.L.; investigation, S.L.; writing—original draft preparation, S.L.; writing—review and editing, E.G. and D.H.; supervision, E.G. and D.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: No new data were created or analyzed in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

Notes

[1] AIS message type 4 is the Base Station Report. Safety-related text is carried by other AIS message types, such as type 12 for addressed safety-related messages and type 14 for safety-related broadcast messages. See Table 1.

[2] AIS message type 21 reports the position and status of an aid to navigation, such as a buoy, beacon, or virtual AtoN, to AIS recipients within range. See Table 1.

[3] AIS message type 9 is the message used by search and rescue aircraft. See Table 1.

[4] Type approval refers to the regulatory certification process through which AIS transponders are tested and validated to ensure compliance with international standards for performance, interoperability, and safety, such as those set by the International Maritime Organization (IMO) and the International Telecommunication Union (ITU). This process helps ensure that devices behave predictably and securely under defined operational conditions.

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